Computational Argumentation Frameworks and Strategyproofness in Grounded Argumentation Mechanisms

Autonomous Systems – Assignment 3 Report

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Abstract

Argumentation frameworks provide a formal way to represent arguments and the conflicts between them. This assignment studies the grounded argumentation mechanism and investigates whether it is strategyproof, i.e., whether an agent can benefit by misreporting the arguments it owns. We implement an argumentation framework engine that supports several standard semantics, including grounded, preferred, stable, complete, and semi-stable semantics. Using this engine, we implement the grounded argumentation mechanism, in which agents report subsets of their privately owned arguments and the mechanism computes the grounded extension of the resulting framework. We examine the theoretical conditions for strategyproofness, with particular focus on the sufficient condition that no two arguments owned by the same agent are connected by an odd-length directed path. We also study examples showing that this condition is sufficient but not necessary. Finally, we evaluate the mechanism through 30,000 randomly generated instances. The mechanism is strategyproof in 7,885 cases (26.28%). The sufficient condition holds in 1,029 cases (3.43%), and all of these cases are strategyproof. However, 6,856 strategyproof cases do not satisfy the condition, showing that the condition identifies only a subset of the strategyproof instances. These results highlight the difference between a sufficient theoretical guarantee and the actual behaviour of the mechanism in practice.

1 Introduction

Abstract Argumentation Frameworks (AFs), introduced by Dung (1995), provide a formal way to represent arguments and the attacks between them. Different argumentation semantics determine which arguments can be accepted. In this assignment, we focus on the *grounded semantics*, which produces a unique set of accepted arguments.

In a multi-agent setting, each agent owns a subset of arguments and can choose which arguments to report. Since the final outcome depends on these reports, an agent may try to hide some arguments to improve its outcome. This raises the question of whether the mechanism is *strategyproof*, meaning that no agent can benefit by misreporting its arguments.

We study the strategyproofness of the grounded argumentation mechanism, examine the theoretical conditions that guarantee it, and evaluate its behaviour through a Python implementation and experiments on randomly generated frameworks.

The main objectives of this assignment are:

- Implement an Abstract Argumentation Framework supporting standard argumentation semantics.
- Develop the grounded argumentation mechanism for multiple agents with privately owned arguments.
- Study strategyproofness and analyse the sufficient condition based on odd-length directed paths.

- Verify the implementation using unit tests and evaluate strategyproofness through experiments on randomly generated frameworks.

2 Background and Theoretical Concepts

2.1 Abstract Argumentation Framework

An Abstract Argumentation Framework (AF), introduced by Dung (1995), is a mathematical model used to represent arguments and the conflicts between them. It is defined as a pair:

$$AF = (A, R)$$

where:

- A is a finite set of arguments.
- $R \subseteq A \times A$ is a set of directed attacks between arguments.

If $(a, b) \in R$, argument a attacks argument b. An attack represents a conflict in which the acceptance of one argument challenges the acceptance of another.

2.2 Conflict-Free Sets and Defence

The basic concepts used to evaluate arguments are:

- **Conflict-Free Set:** A set of arguments is conflict-free if no argument in the set attacks another argument in the same set.
- **Defence:** An argument a is defended by a set $S \subseteq A$ if every argument attacking a is itself attacked by at least one argument in S.
- **Acceptability:** An argument is acceptable with respect to a set S if it is defended by S against all possible attacks.

These concepts form the foundation for determining which arguments can be accepted in an argumentation framework.

2.3 Argumentation Semantics

Argumentation semantics define different criteria for selecting acceptable sets of arguments.

- **Conflict-Free:** A set containing no arguments that attack one another.
- **Admissible:** A conflict-free set that defends all its arguments against every attack.
- **Complete:** An admissible set that contains every argument it defends.
- **Grounded:** The least complete extension under set inclusion. It is unique for every argumentation framework and represents a cautious acceptance of arguments.
- **Preferred:** A maximal admissible set, meaning that no additional arguments can be added without violating admissibility.
- **Stable:** A conflict-free set that attacks every argument outside the set.
- **Semi-Stable:** A complete extension whose range is maximal under set inclusion. The range consists of the arguments in the extension and the arguments they attack.

Different semantics may produce different accepted sets for the same framework. In this assignment, we primarily focus on grounded semantics because it always produces a unique extension.

2.4 Characteristic Function and Grounded Extension

The characteristic function is used to determine which arguments are acceptable with respect to a given set of arguments.
For an argumentation framework $AF = (A, R)$, the characteristic function F is defined as:

$$F(S) = \{a \in A \mid a \text{ is defended by } S\}$$

In other words, $F(S)$ contains all arguments that are defended against every attack by arguments in the set S.
The grounded extension is obtained by repeatedly applying the characteristic function, starting with the empty set:

$$S_0 = \emptyset$$

$$S_{i+1} = F(S_i)$$

This process continues until the set stops changing:

$$S_{k+1} = S_k$$

The resulting set S_k is called the **grounded extension**. It is the least fixed point of the characteristic function and represents the arguments accepted under grounded semantics.
Since the process starts with the empty set and adds arguments that become defended, it provides a systematic way to compute the grounded extension. The process always terminates for a finite argumentation framework.

2.5 Caminada's Three-Valued Labelling

Caminada's labelling approach classifies every argument in an argumentation framework into one of three categories:

- **In:** The argument is accepted.
- **Out:** The argument is rejected because it is attacked by an accepted argument.
- **Undecided:** The argument can neither be accepted nor rejected based on the available information.

A complete labelling follows these rules:

- An argument is labelled **In** if all its attackers are labelled Out.
- An argument is labelled **Out** if at least one of its attackers is labelled In.
- An argument is labelled **Undecided** if it has no attacker labelled In and at least one attacker labelled Undecided.

Under grounded semantics, the grounded labelling is the least complete labelling. Its In-labelled arguments form the grounded extension, while the Out and Undecided arguments represent rejected and unresolved arguments, respectively. This approach provides a clear classification of every argument and is useful for understanding and implementing argumentation semantics.

2.6 Preferred Acceptance Classification

A framework can have multiple preferred extensions. Based on these extensions, each argument can be classified into one of three categories:

- **Skeptically Accepted:** An argument is skeptically accepted if it belongs to every preferred extension.
- **Credulously Accepted:** An argument is credulously accepted if it belongs to at least one preferred extension, but not necessarily all of them.
- **Rejected:** An argument is rejected if it does not belong to any preferred extension.

These classifications help distinguish arguments that are consistently accepted from those whose acceptance depends on the selected preferred extension.
Unlike grounded semantics, which always produces a unique extension, preferred semantics may produce multiple extensions. This makes preferred acceptance useful for analyzing different possible outcomes of an argumentation framework.

3 Grounded Argumentation Mechanism

In a multi-agent argumentation framework, the set of arguments is divided among different agents. Each agent privately owns a subset of arguments and can choose which of these arguments to report.
Let N be the set of agents and A_i be the set of arguments owned by agent i. The argument sets form a partition of A, meaning that every argument belongs to exactly one agent.
Each agent reports a subset $A'_i \subseteq A_i$. The mechanism combines the reported arguments and considers the attacks between them to construct the reported framework. It then computes the grounded extension of this framework.
The final grounded extension determines which reported arguments are accepted. Thus, each agent's outcome depends not only on its own report but also on the reports submitted by the other agents.

3.1 Utility and Strategyproofness

Each agent evaluates the outcome based on how many of its reported arguments are accepted in the grounded extension. The utility of agent i is therefore defined as:

$$u_i(A'_i, A'_{-i}) = \left|A'_i \cap GE(A')\right|$$

where A'_i is the agent's reported set, A'_{-i} represents the reports of all other agents, and $GE(A')$ is the grounded extension of the framework formed using all reported arguments.
A mechanism is called *strategyproof* if no agent can increase its utility by misreporting its arguments while the reports of all other agents remain unchanged.
Formally, for every agent i, every truthful argument set A_i, every alternative report $A'_i \subseteq A_i$, and every fixed set of reports by the other agents, the following condition must hold:

$$u_i(A_i, A'_{-i}) \geq u_i(A'_i, A'_{-i})$$

Thus, truthful reporting should always provide an outcome at least as good as any alternative report.

4 Theoretical Results

4.1 Theorem 5.3: Characterisation of Strategyproofness

Theorem 5.3 establishes that truthful reporting is a weakly dominant strategy for an agent if and only if truthful reporting is a best response against every possible combination of reports from the other agents.
In other words, an agent should not be able to improve its utility by changing its report, regardless of what the other agents report. This provides a formal basis for checking whether a mechanism is strategyproof.

4.2 Theorem 5.4: Sufficient Condition for Strategyproofness

Theorem 5.4 provides a sufficient condition for the grounded argumentation mechanism to be strategyproof. It states that if no two arguments owned by the same agent are connected by a directed path of odd length, then the mechanism is strategyproof.
A directed path of odd length contains an odd number of attack edges. The absence of such paths between arguments owned by the same agent guarantees that hiding arguments cannot improve the agent's utility.
However, this is a sufficient condition, not a necessary one. Therefore, a mechanism may still be strategyproof even when the condition is not satisfied. We investigate this distinction through the examples and experiments discussed in the following sections.

4.3 Example 5.10: A Strategyproofness Violation

Consider an argumentation framework with five arguments:

$$A = \{a_1, a_2, a_3, a_4, a_5\}$$

The attack relations are:

$$R = \{(a_1, a_2), (a_2, a_3), (a_3, a_4), (a_3, a_5)\}$$

The arguments are distributed among three agents:

$$A_x = \{a_1, a_4, a_5\}, \quad A_y = \{a_2\}, \quad A_z = \{a_3\}$$

When all agents report truthfully, the grounded extension is:

$$GE(A) = \{a_1, a_3\}$$

Agent x has only one accepted argument, a_1, so its utility is:

$$u_x = 1$$

Now, suppose agent x hides argument a_1. The resulting grounded extension becomes:

$$GE(A') = \{a_2, a_4, a_5\}$$

In this case, both a_4 and a_5 are accepted, giving agent x a utility of:

$$u_x = 2$$

Since hiding an argument increases its utility from 1 to 2, the mechanism is not strategyproof in this example.

4.4 Example 5.11: Strategyproofness Without the Sufficient Condition

Consider an extended framework with an additional argument a_6 and an attack relation (a_6, a_3). The argument set and ownership are now:

$$A = \{a_1, a_2, a_3, a_4, a_5, a_6\}$$

$$A_x = \{a_1, a_4, a_5, a_6\}, \quad A_y = \{a_2\}, \quad A_z = \{a_3\}$$

When all agents report truthfully, the grounded extension is:

$$GE(A) = \{a_1, a_4, a_5, a_6\}$$

Therefore, agent x receives a utility of:

$$u_x = 4$$

If agent x hides a_1, the grounded extension becomes:

$$GE(A') = \{a_2, a_4, a_5, a_6\}$$

Agent x then receives a utility of:

$$u_x = 3$$

Thus, hiding a_1 does not improve agent x's utility.
Unlike Example 5.10, this framework still satisfies strategyproofness, even though the sufficient condition in Theorem 5.4 is violated. For instance, the path $a_1 \to a_2 \to a_3 \to a_4$ has odd length and connects two arguments owned by agent x. This example demonstrates that the condition in Theorem 5.4 is sufficient but not necessary for strategyproofness.

5 Experimental Setup and Methodology

To evaluate the strategyproofness of the grounded argumentation mechanism, we conduct experiments on randomly generated argumentation frameworks.

5.1 Experimental Setup

The experiments use the following configuration:

- **Number of agents:** 3
- **Number of arguments:** 6 to 8
- **Attack probabilities:** 0.15, 0.20, 0.25, 0.30, 0.35, and 0.40
- **Instances per probability:** 5,000
- **Total instances:** 30,000

5.2 Evaluation Methodology

For each generated framework, we evaluate whether the grounded argumentation mechanism is strategyproof by comparing an agent's utility under truthful reporting with its utility under alternative reports.
We also check whether the sufficient condition stated in Theorem 5.4 holds for each instance. Based on these two checks, the instances are classified into three categories: strategyproof instances satisfying the condition, strategyproof instances violating the condition, and non-strategyproof instances.
This evaluation allows us to measure how frequently the mechanism is strategyproof and how accurately the theoretical condition identifies strategyproof instances.

6 Experimental Results and Analysis

The experiments evaluate the relationship between the strategyproofness of the grounded argumentation mechanism and the sufficient condition stated in Theorem 5.4.

6.1 Results

Table 1 presents the results for each attack probability. The columns distinguish strategyproof instances covered by the sufficient condition from strategyproof instances not covered by it.

Table 1: Strategyproofness and Theorem 5.4 condition across attack probabilities

Attack probability	Strategyproof	Condition holds	SP, condition fails	Non-strategyproof
0.15	2637 (52.74%)	545 (10.90%)	2092 (41.84%)	2363 (47.26%)
0.20	1899 (37.98%)	298 (5.96%)	1601 (32.02%)	3101 (62.02%)
0.25	1314 (26.28%)	127 (2.54%)	1187 (23.74%)	3686 (73.72%)
0.30	874 (17.48%)	43 (0.86%)	831 (16.62%)	4126 (82.52%)
0.35	650 (13.00%)	10 (0.20%)	640 (12.80%)	4350 (87.00%)
0.40	511 (10.22%)	6 (0.12%)	505 (10.10%)	4489 (89.78%)
Total	**7885 (26.28%)**	**1029 (3.43%)**	**6856 (22.85%)**	**22115 (73.72%)**

6.2 Analysis

Out of 30,000 instances, 7,885 (26.28%) were strategyproof. The sufficient condition held in 1,029 instances (3.43%), and all of these were strategyproof. No instance satisfied the condition while violating strategyproofness.

However, 6,856 instances were strategyproof even though they did not satisfy the condition. These account for 86.95% of all strategyproof instances. This shows that the condition guarantees strategyproofness but identifies only a subset of strategyproof frameworks.

The proportion of strategyproof instances also decreases as the attack probability increases, from 52.74% at $p = 0.15$ to 10.22% at $p = 0.40$. This trend is observed in the tested settings and suggests that denser attack relations are associated with fewer strategyproof instances.

7 Complexity Analysis

The computational complexity of the implementation depends on the number of arguments $n = |A|$ and the number of attack relations $m = |R|$.
The main operations have the following complexity:

- **Conflict-Free Check:** Checking whether a set contains any internal attack takes $O(m)$ time.
- **Defence Check:** Checking whether an argument is defended requires examining its attackers and the corresponding attacks. A straightforward implementation takes up to $O(nm)$ time.
- **Grounded Extension:** The characteristic function is repeatedly applied until a fixed point is reached. Since at most n arguments can be added, the overall computation is polynomial.
- **Preferred and Stable Extensions:** Computing these extensions may require examining multiple subsets of arguments. In the worst case, this involves exponential time in n.

The grounded semantics is computationally more manageable than preferred or stable semantics because it produces a unique extension through a fixed-point computation.

8 Conclusion

In this assignment, we implemented an Abstract Argumentation Framework engine and studied the strategyproofness of the grounded argumentation mechanism. We examined the theoretical condition based on odd-length directed paths and analysed examples demonstrating that this condition is sufficient but not necessary.

Our experiments on 30,000 randomly generated instances found that 7,885 (26.28%) were strategyproof. The condition in Theorem 5.4 guaranteed strategyproofness in all the instances where it held. However, 6,856 strategyproof instances did not satisfy this condition.

These results demonstrate that while the theoretical condition provides a guarantee of strategyproofness, it does not capture all cases in which truthful reporting is beneficial. The implementation and experiments together provide a practical evaluation of the grounded argumentation mechanism and its strategic properties.

References

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